This is a static version of the slides — some animations are omitted for clarity.

NTNU

Norwegian University of Science and Technology

Understanding what matters: UQSA in cardiovascular models

Concepts, methods, and applications

Leif Rune Hellevik

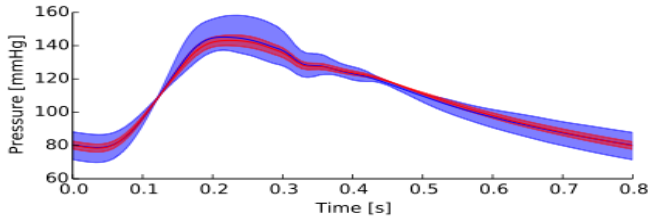
CVBL UMich-Seminar April 23th, 2026

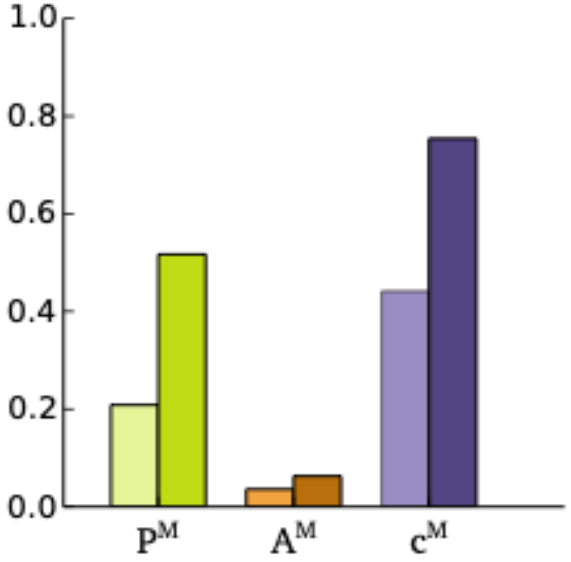
The problem is not uncertainty

- Models are always uncertain
  - Imperfect representations (physiological systems)
  - uncertain inputs and states
- We cannot remove all uncertainty
- Key questions
  - How uncertain are the predictions?
  - What part of it actually matters?

What you will get from this talk

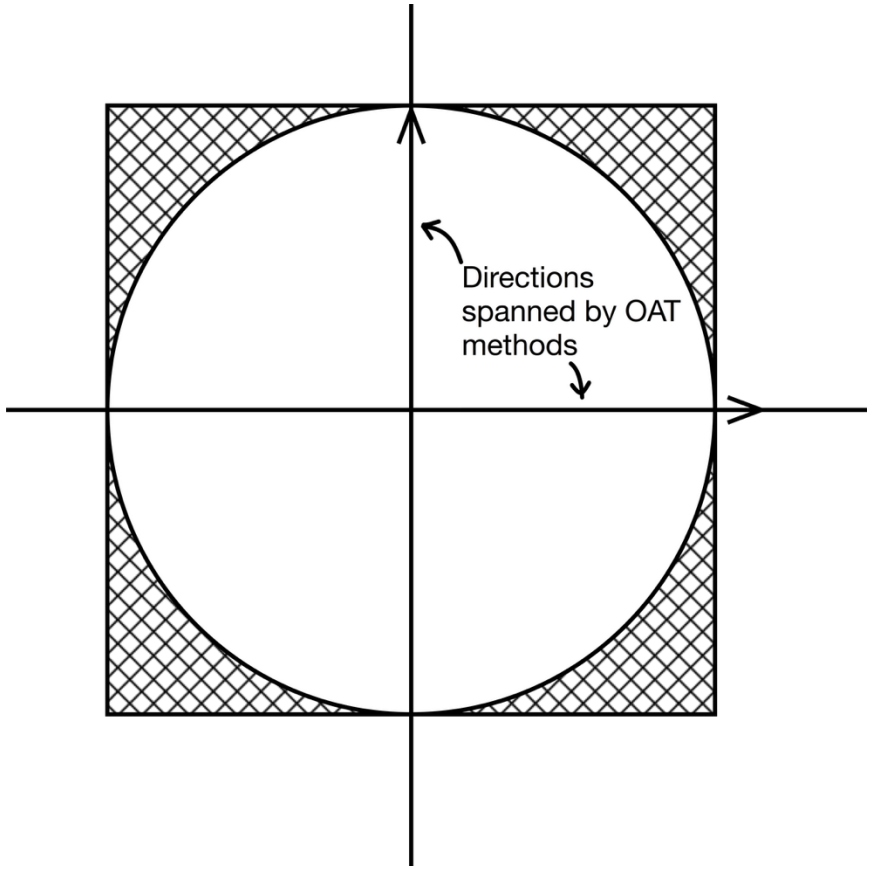
- Understanding of
  - uncertainty quantification (UQ)
  - sensitivity analysis (SA)
- Variance-based methods (Sobol indices)
- How UQSA is used in cardiovascular modeling





Why intuition fails in complex models

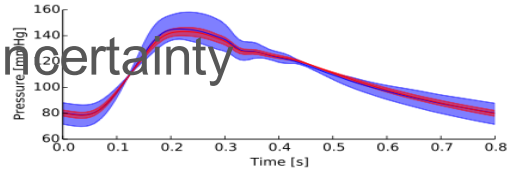
- One-factor-at-a-time (OAT) thinking
  - Local sensitivity (derivatives)
- Problem:
  - Only valid near one point
  - Misses interactions
  - Explores almost none of the input space
- Most of the model behaviour is never seen
- So we need methods that
  - explore the full input space
  - identify what matters



A framework for UQ and sensitivity analysis

Propagation

Input uncertainty → Model → Output uncertainty



uncertainty quantification (UQ)

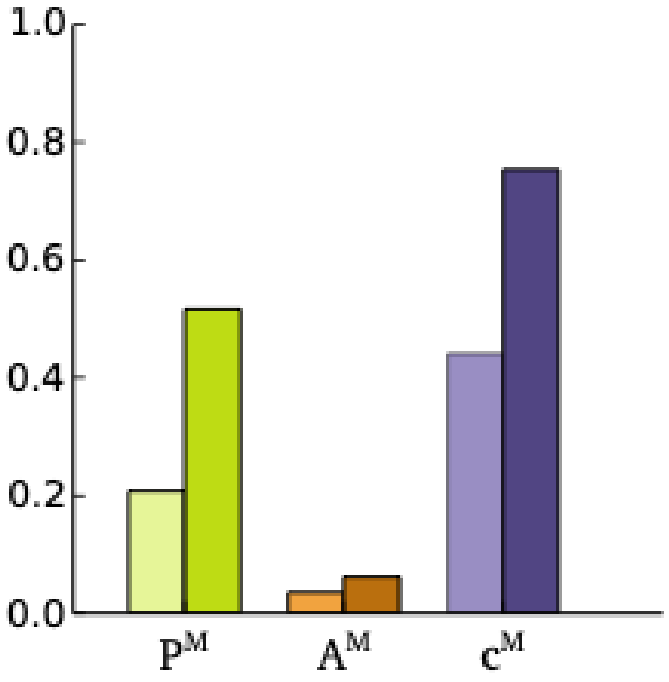
model structure
level of detail

prioritisation of parameters
fixation and/or accuracy

sensitivity analysis (SA)

What sensitivity analysis (SA) is used for

- What matters most?
  - Which inputs drive output uncertainty
- What can be ignored?
  - Identify non-influential inputs that can be fiexed
- How the model behaves
  - Nonlinear effects and interactions
- What to do about it?
  - Focus where it matters

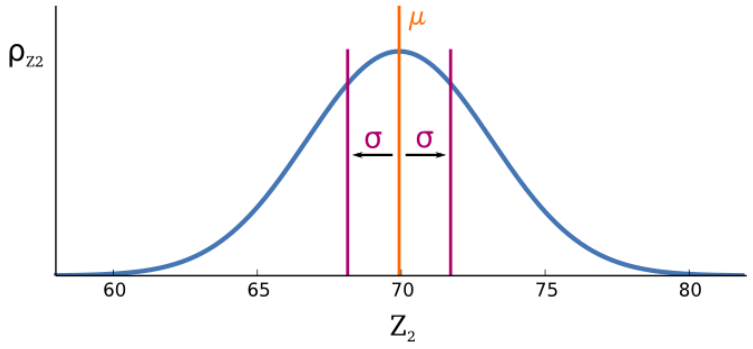


Uncertainty quantification

$$\mathbf{Z} = \{Z_1, \ldots, Z_D\} \rightarrow f(\mathbf{Z}) \rightarrow \boxed{\mathbf{Y}}$$

random variables

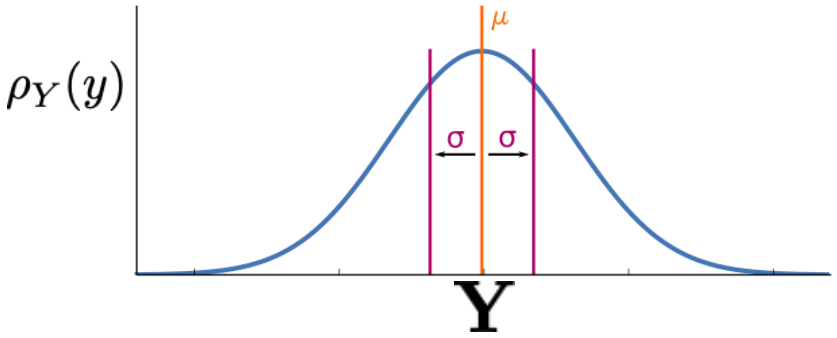
known distributions



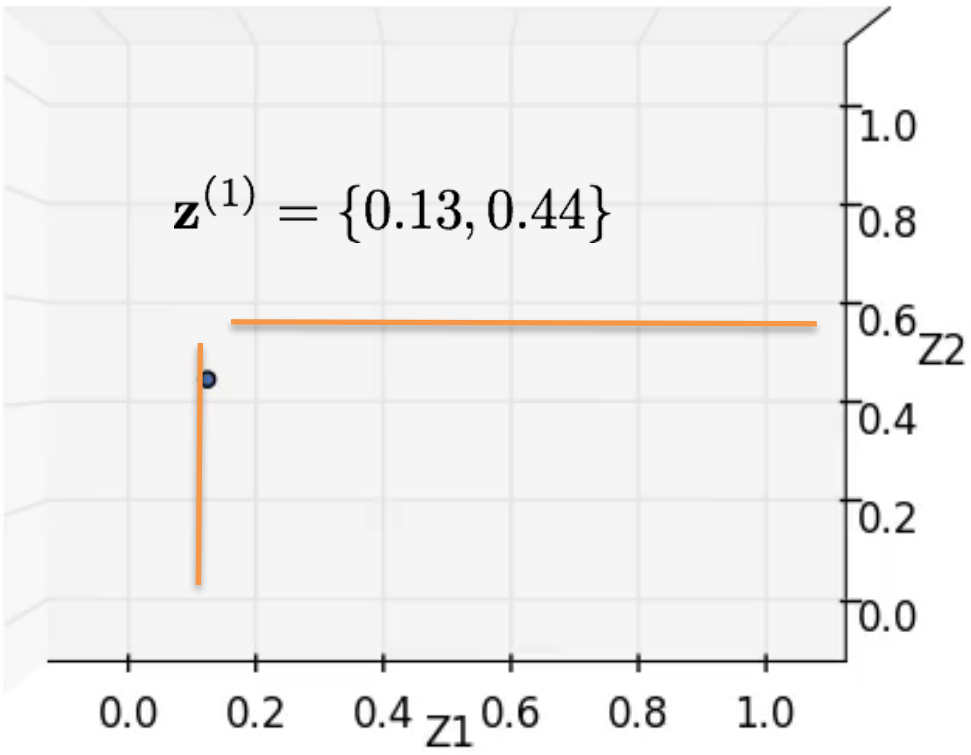
random variable

unknown distribution

?



Sampling



random variable vector

$$\mathbf{Z} = \{Z_1, Z_2\}$$

$$Z_1 \sim \mathcal{U}(0,1), \; Z_2 \sim \mathcal{U}(0,1)$$

set of samples

$$\left\{\mathbf{z}^{(s)}\right\}_{s=1}^{N_s}$$

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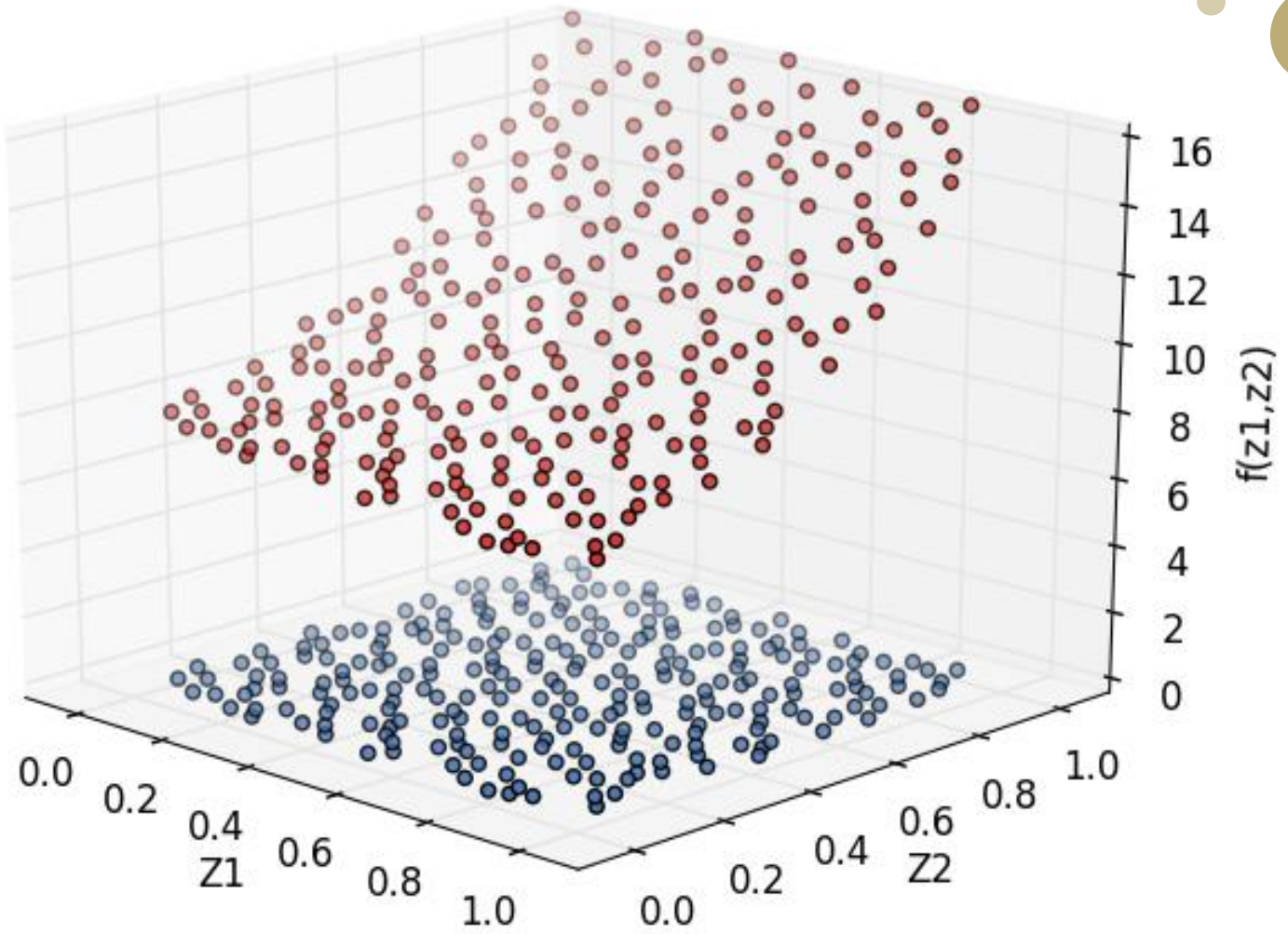
Model evaluation



$$\left\{\mathbf{z}^{(s)}\right\}_{s=1}^{N_s} \rightarrow \mathbf{y}^{(s)} = f\left(\mathbf{z}^{(s)}\right) \rightarrow \left\{\mathbf{y}^{(s)}\right\}_{s=1}^{N_s}$$

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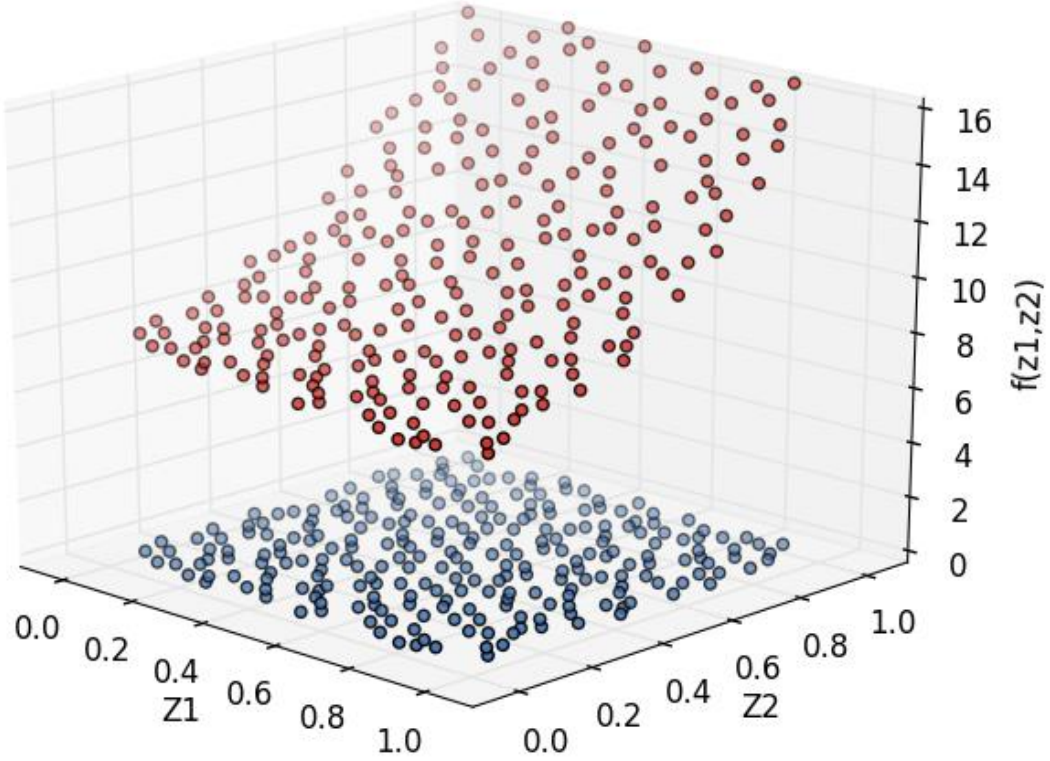
Monte carlo method

- Requires many model evaluations
- Uses samples directly to compute statistics
- Extra samples needed for Sobol indices

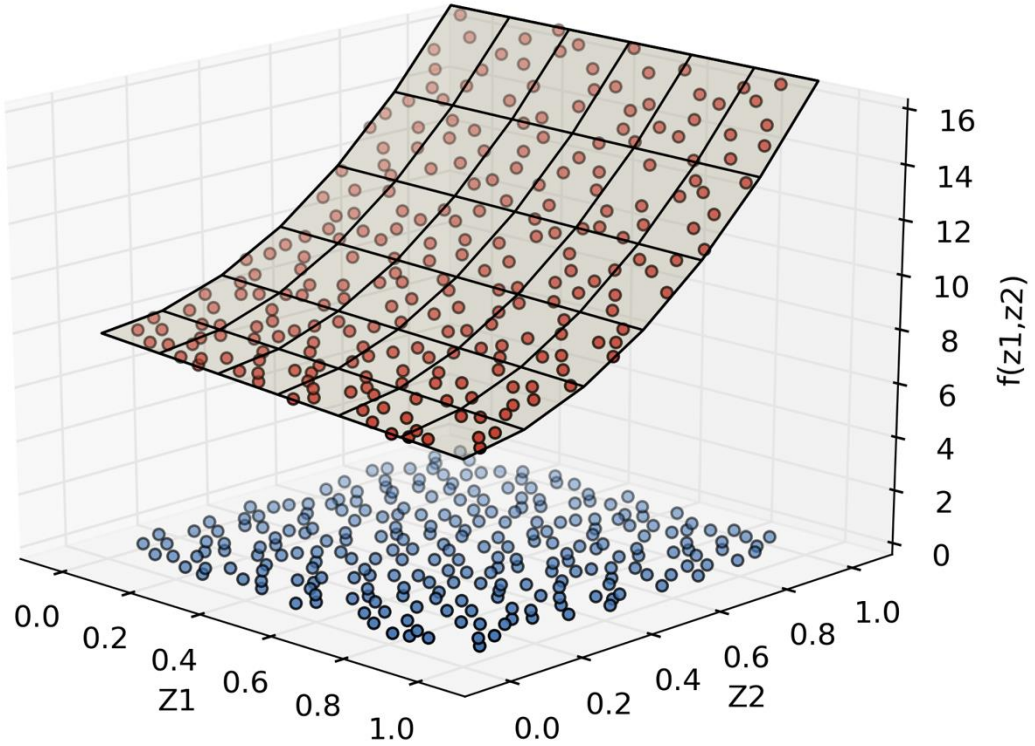


$$\mathbb{E}\left[\mathbf{Y}\right] \approx \frac{1}{N_{\mathrm{s}}} \sum_{s=1}^{N_{\mathrm{s}}} \mathbf{y}^{(s)}$$

Polynomial chaos method

- Approximation with a spectral expansion
- Convenient computation of statistics
- No extra samples needed for Sobol indices

$$\mathbb{E}[\mathbf{Y}] \approx c_0$$

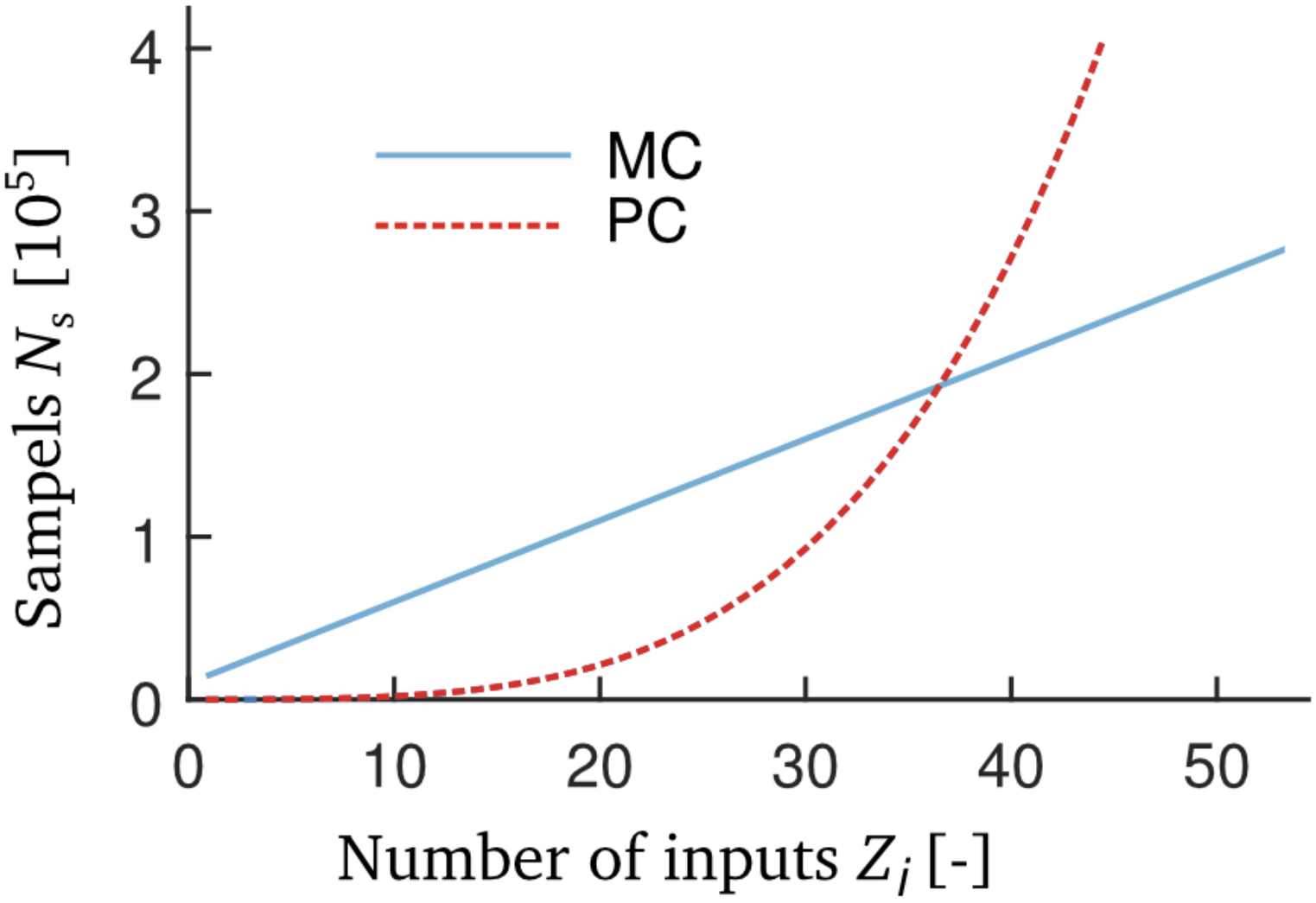


expansion coefficients

$$f(\mathbf{Z}) \approx \hat{f}(\mathbf{Z}) = \sum_{n=0}^{N_p} c_n \Phi_n(\mathbf{Z})$$

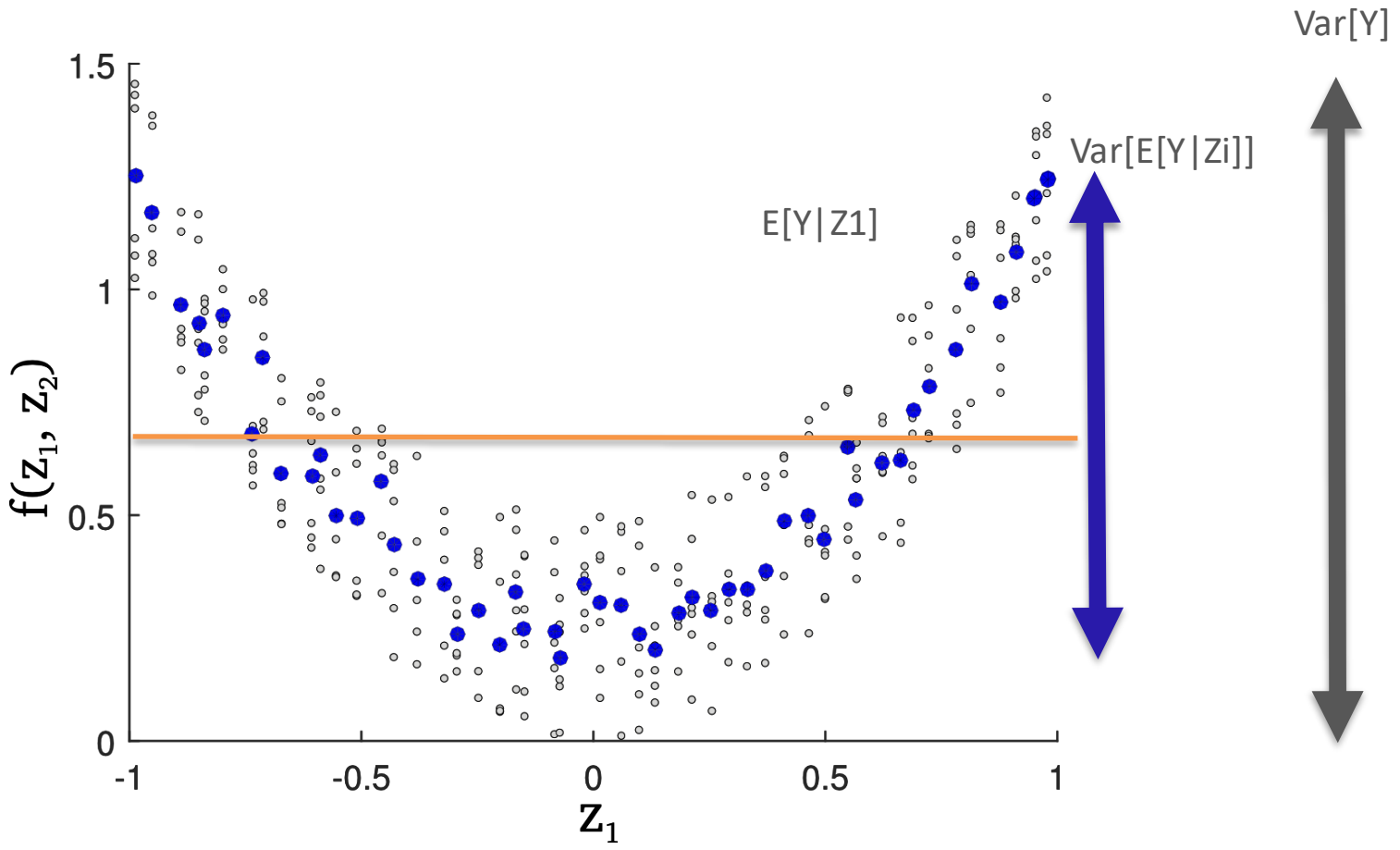
orthogonal polynomials

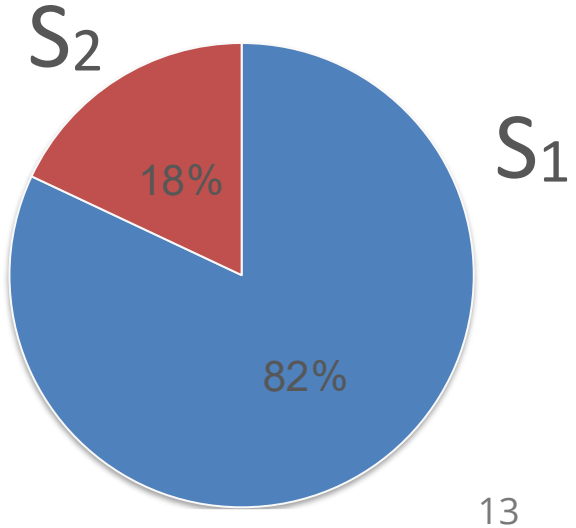
Monte carlo vs polynomial chaos



Sensitivity analysis

First order sensitivity indices $S_i = \frac{\mathrm{Var}\left[\ \mathbb{E}\left[\mathbf{Y}|Z_i\right]\ \right]}{\mathrm{Var}\left[\mathbf{Y}\right]}$





Parameter prioritisation

Total sensitivity indices

Variance Decomposition

$$\mathrm{Var}\left[\mathbf{Y}\right] = V_1 + V_2 + V_3 + V_{12} + V_{13} + V_{23} + V_{123}$$

divide by $\mathrm{Var}\left[\mathbf{Y}\right]$

$$1 = S_1 + S_2 + S_3 + S_{12} + S_{13} + S_{23} + S_{123}$$

$$S_{T1} = S_1 + S_{12} + S_{13} + S_{123}$$

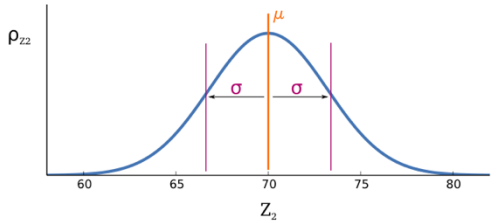
- **Total index = main effect + all interactions**
- **Key for identifying interactions and parameter fixing**

6 steps to UQ and SA

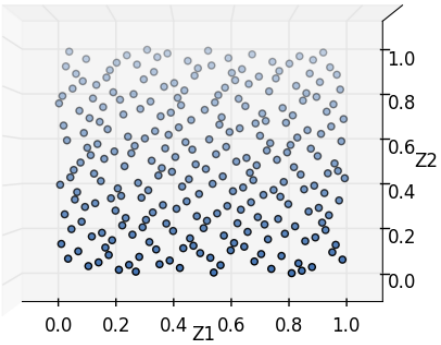
1. Identify the output of interest

$$\mathrm{FFR} \in [0,1] \begin{cases} \text{non/significant} & \text{for FFR} > 0.8 \\ \text{sigificant} & \text{for FFR} < 0.8 \end{cases}$$

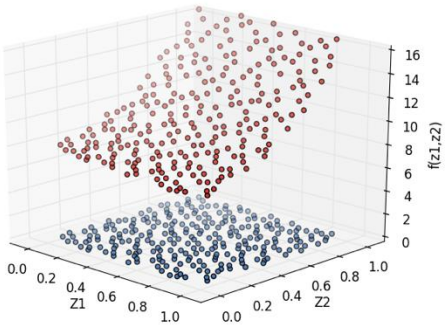
2. Identify input distributions



3. Sample the input space

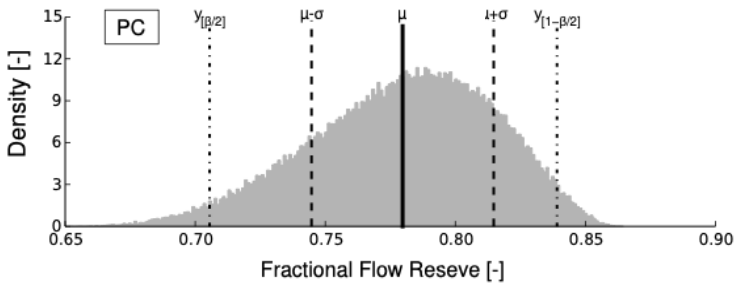


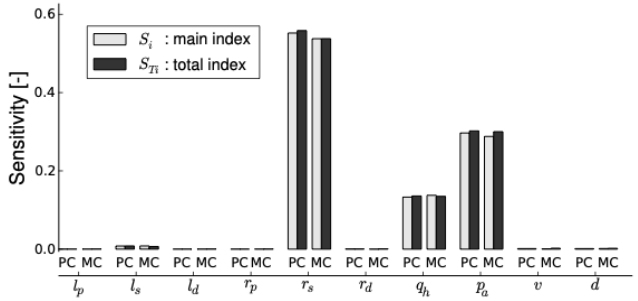
4. Evaluate the model for all samples



5. Perform UQ and SA

6. Assess convergence

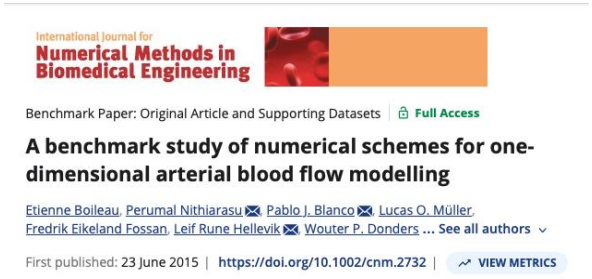




STARFiSh

STochastic ARterial Flow Simulations

NTNU


International Journal for
Numerical Methods in Biomedical Engineering

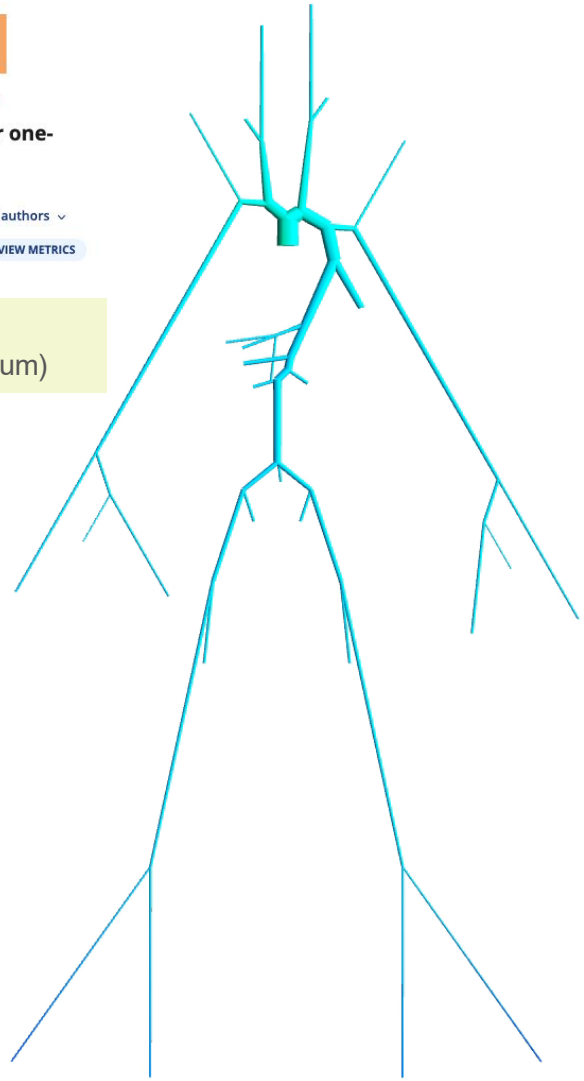
Benchmark Paper: Original Article and Supporting Datasets | Full Access

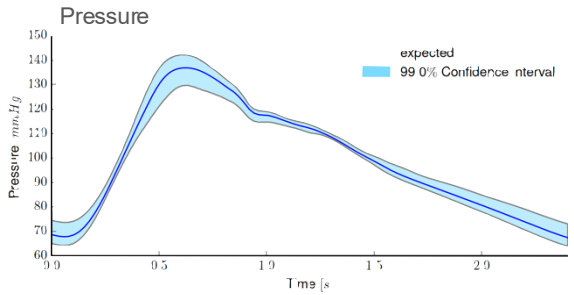
A benchmark study of numerical schemes for one-dimensional arterial blood flow modelling

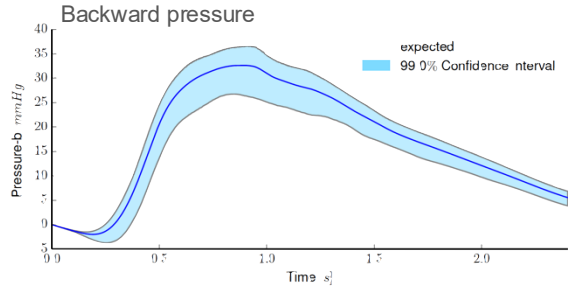
Etienne Boileau, Perumal Nithiarasu, Pablo J. Blanco, Lucas O. Müller, Fredrik Eikeland Fossan, Leif Rune Hellevik, Wouter P. Donders ... See all authors

First published: 23 June 2015 | https://doi.org/10.1002/cnm.2732 | VIEW METRICS

Benchmark validated
(DG, FEM, FV, MacCormack, trapezium)







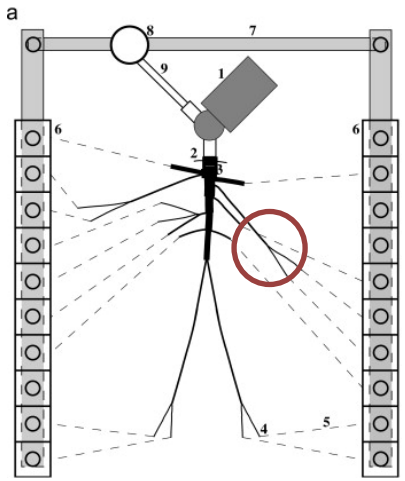
$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = 0$$

$$\frac{\partial Q}{\partial t} + \delta \frac{\partial}{\partial x}\left(Q^2/A\right) = -\frac{A}{\rho}\frac{\partial p}{\partial x} + \tau_w$$

$$p(A) = p_s e^{\beta(A/A_s - 1)}$$

$$u(x,t;z) \simeq \tilde{u}(x,t;z) = \sum_{i=0}^{M} c_i(x,t)\,\phi_i(z)$$

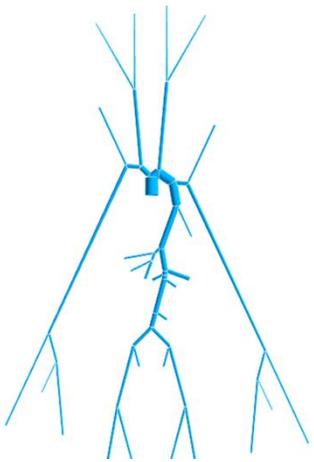
Effects of uncertainties in wall models

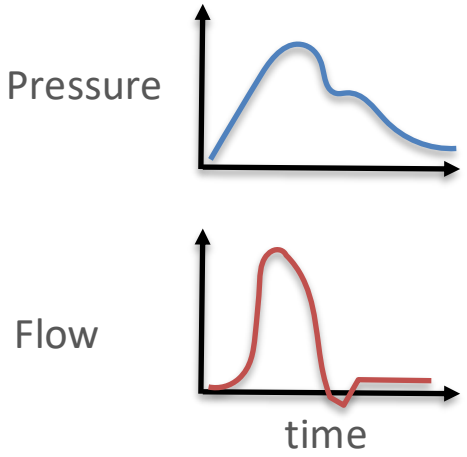


wave speed c^M

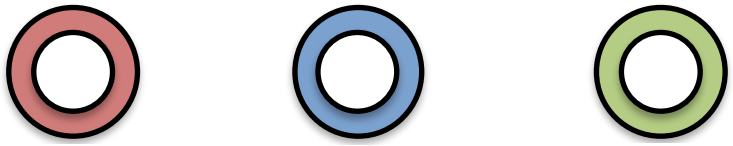
brachial pressure P^M

brachial area A^M





Arterial wall models



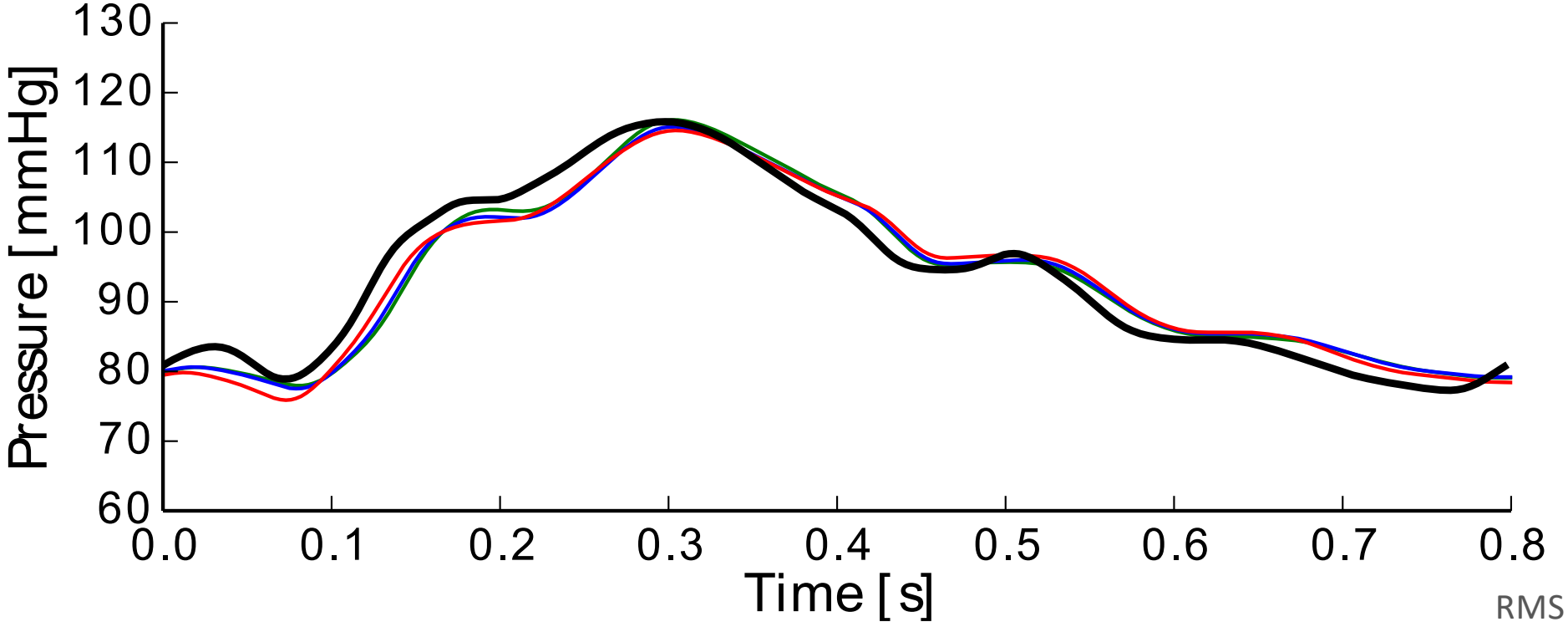
quadratic logarithmic arctan

Journal of Biomechanics
Volume 50, 4 January 2017, Pages 188-194

Effects of arterial wall models and measurement uncertainties on cardiovascular model predictions

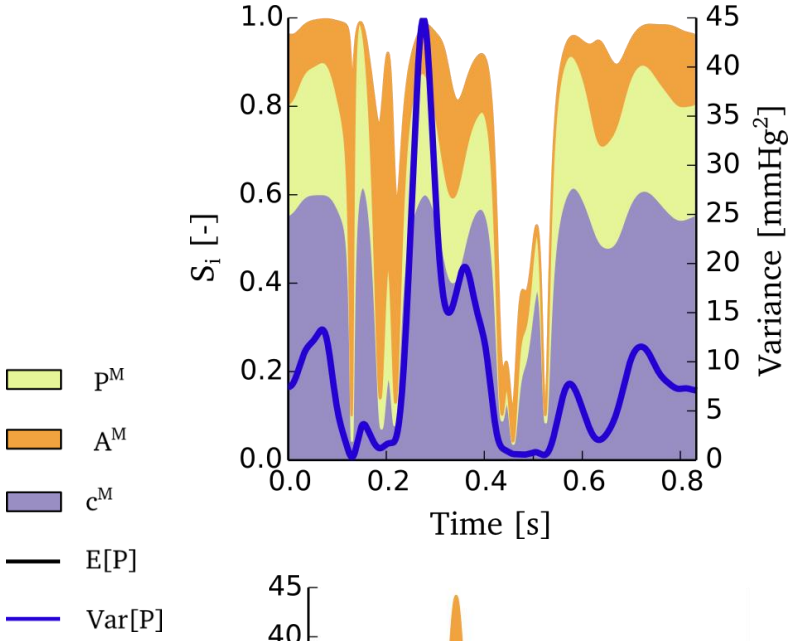
V.G. Eck, J. Sturdy, L.R. Hellevik

Uncertainty quantification

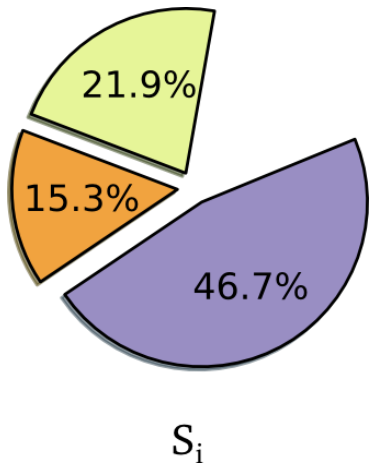


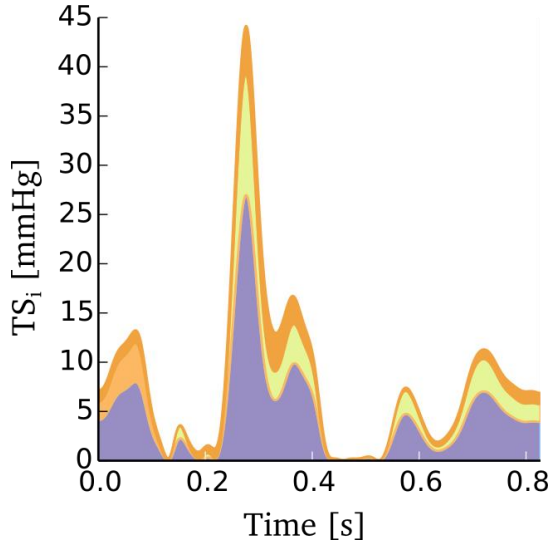
	RMS error of E(Z) and data	Prediction interval average
quadratic	2.6 %	8.7 mmHg
log.	2.5 %	9.9 mmHg
arctan	2.4 %	12.0 mmHg

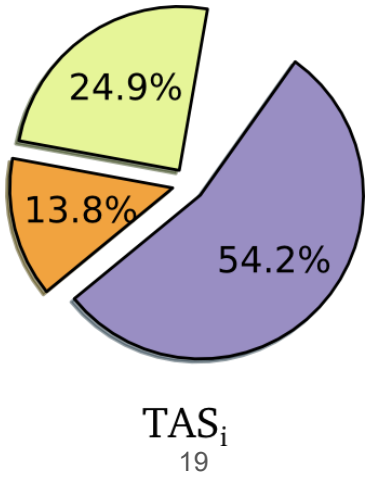
Time averaged sensitivity indices



$$\frac{\int S_i(t)V_i(t)\,dt}{\int V_i(t)\,dt}$$







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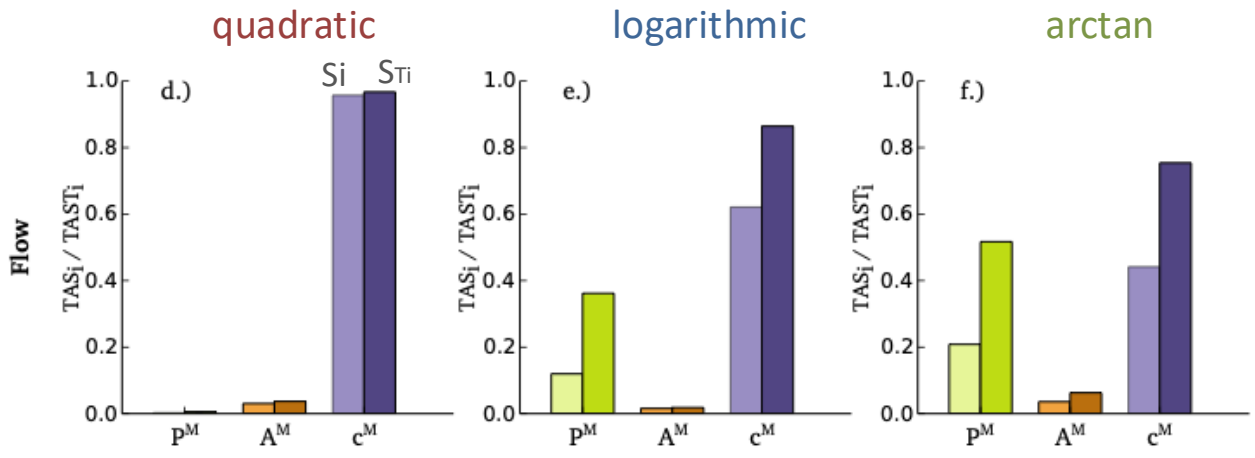
SA of the wall models



	RMS error of E(Z) and data	Prediction interval average
quadratic	2.6 %	8.7 mmHg
log.	2.5 %	9.9 mmHg
arctan	2.4 %	12.0 mmHg

The key takeaway:

- Novel time-averaged Sobol-indices valuable for time-varying QoIs
- Wall model choice changes both uncertainty and its drivers
- Low RMS does not guarantee low predictive uncertainty

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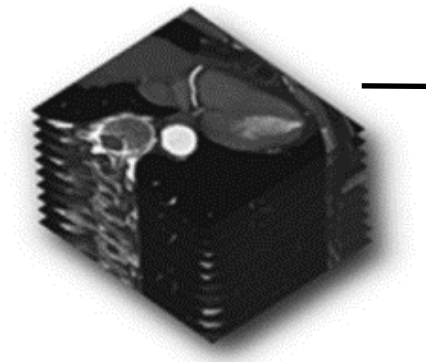
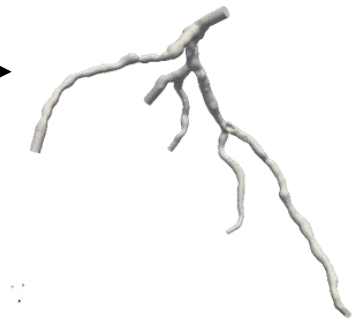
Patient specific FFR prediction

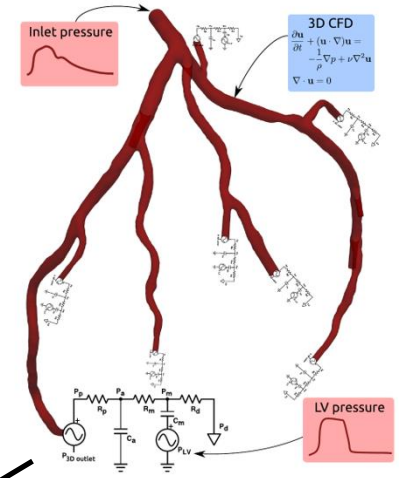
Coronary CT



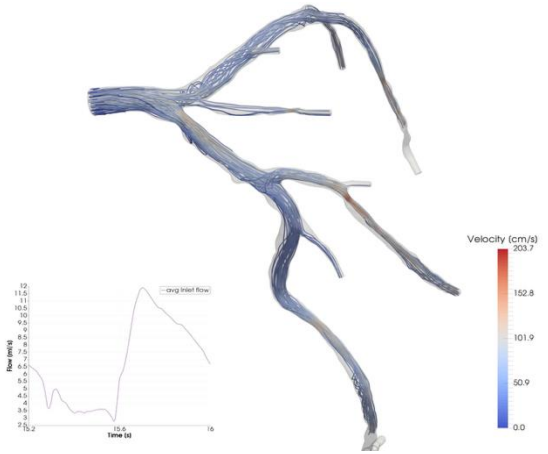
Anatomical model



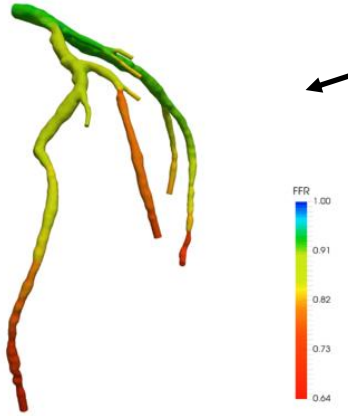
Physiological model



Flow simulations



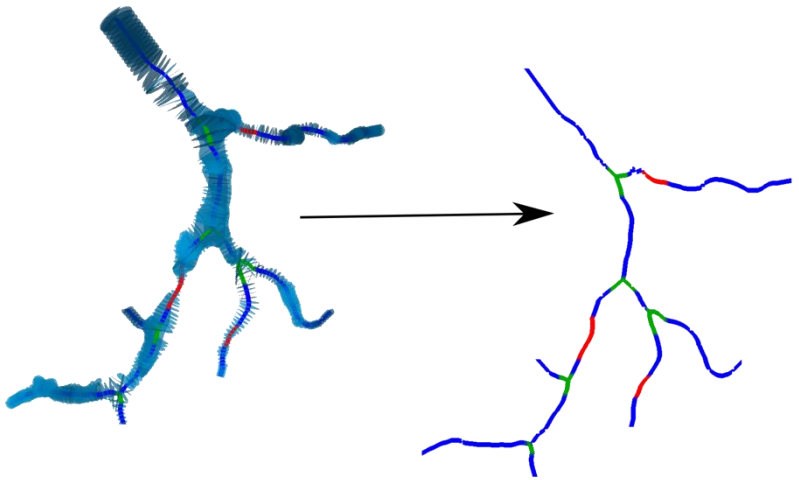
Computational FFR



Reduced order model (ROM) for FFR-prediction

- Aim: Reduced order model for

$$\mathbf{u} \cdot \boldsymbol{\nabla}\mathbf{u} = -\frac{1}{\rho}\boldsymbol{\nabla}p + \frac{\mu}{\rho}\nabla^2\mathbf{u}$$



- The domain is separated into
  - Healthy/straight sections
  - Stenotic sections
  - Bifurcations

$$\frac{\partial(\frac{Q^2}{A})}{\partial x} = \frac{-A}{\rho}\frac{\partial p}{\partial x} - \frac{2(\gamma+2)\mu\pi}{\rho}\frac{Q}{A}$$

$$\Delta p = \frac{K_v\mu}{A_0 D_0}Q + \frac{K_t}{2A_0^2}(\frac{A_0}{A_s} - 1)^2 Q^2$$

$$p_n + \frac{\rho}{2}\frac{{Q_n}^2}{A_n} = constant$$

Cardiovascular Engineering and Technology
December 2018, Volume 9, Issue 4, pp 597–622 | Cite as

Uncertainty Quantification and Sensitivity Analysis for Computational FFR Estimation in Stable Coronary Artery Disease

Authors and affiliations

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251 Downloads 3 Citations

Article
First Online: 31 October 2018

- 13 patients
- 25 FFR measurements
- one flow configuration

International Journal for Numerical Methods in Biomedical Engineering

SPECIAL ISSUE ARTICLE | Open Access

Impact of baseline coronary flow and its distribution on fractional flow reserve prediction

Lucas O. Müller, Fredrik E. Fossan, Anders T. Bråten, Arve Jørgensen, Rune Wiseth, Leif R. Hellevik

First published: 09 August 2019 | https://doi.org/10.1002/cnm.3246

VIEW METRICS

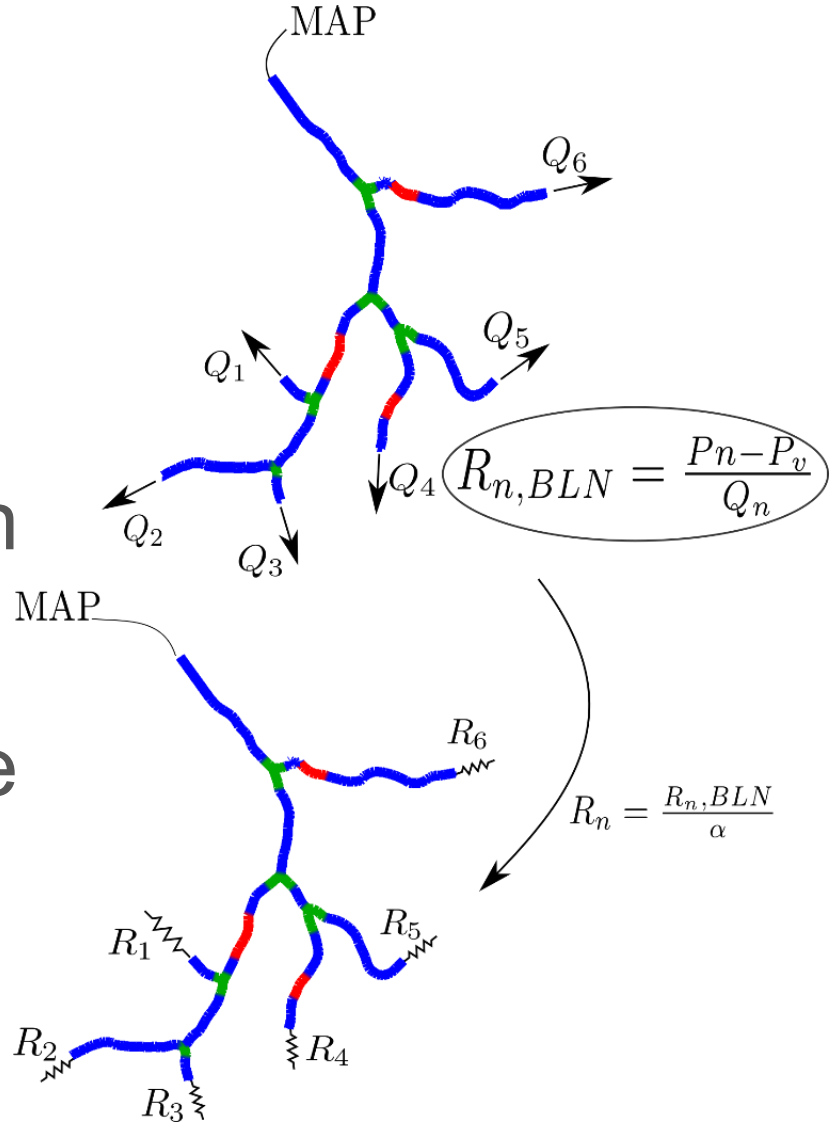
- 63 patients
- 105 FFR measuremer
- 3 flow configurations

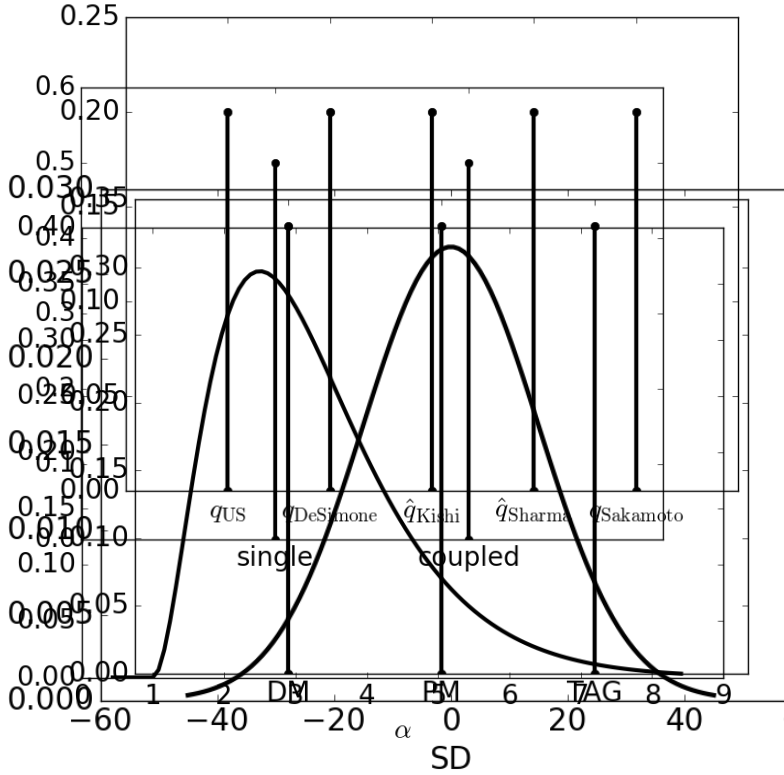
ROM vs CFD

Comparison type	Flow distribution	a	b	r	$FFR_A - FFR_B$	Acc	Sen	Spe	PPV	NPV
RO vs 3D	DM	1.06	-0.04	0.98	-0.01 (0.03)	97.14	95.65	97.56	91.67	98.77
	PM	1.06	-0.04	0.98	-0.01 (0.03)	95.24	84.62	98.73	95.65	95.12
	TAG	1.08	-0.05	0.98	-0.01 (0.04)	94.29	80.00	100.00	100.00	92.59
RO vs invasive measurement	DM	0.64	0.34	0.58	-0.05 (0.13)	81.90	57.58	93.06	79.17	82.72
	PM	0.63	0.34	0.57	-0.04 (0.14)	80.95	54.55	93.06	78.26	81.71
	TAG	0.77	0.21	0.62	-0.03 (0.14)	83.81	60.61	94.44	83.33	83.95
3D vs invasive measurement	DM	0.62	0.35	0.60	-0.04 (0.12)	80.95	54.55	93.06	78.26	81.71
	PM	0.64	0.33	0.62	-0.03 (0.12)	81.90	60.61	91.67	76.92	83.54
	TAG	0.75	0.22	0.66	-0.01 (0.12)	83.81	69.70	90.28	76.67	86.67

SA for ROM inputs

- Baseline flow
- Branch treatment
- Flow distribution
- α
- Stenosis degree



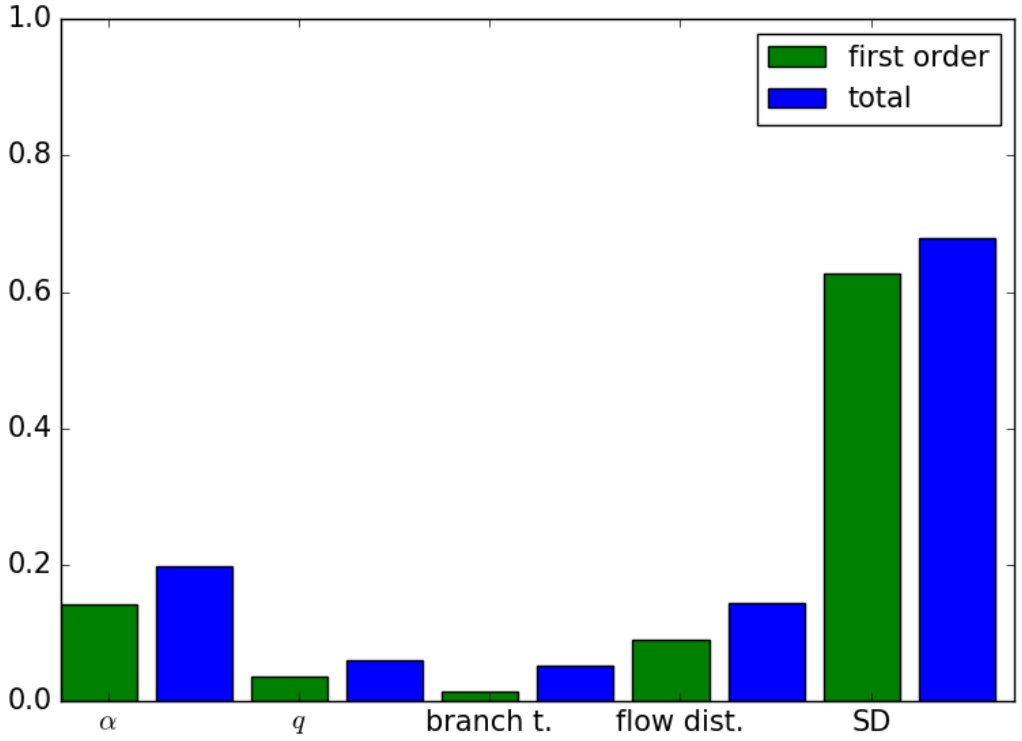


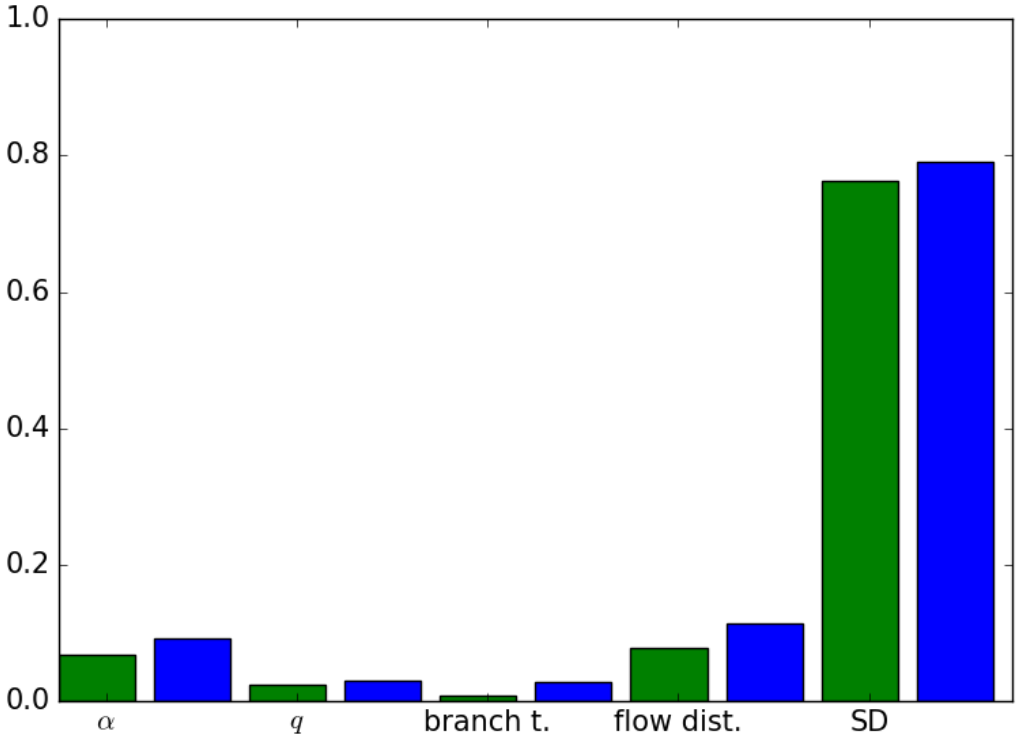
SA of ROM-estimates of FFR

63 patients, 105 FFR measurements

$$S_{i,averaged} = \frac{1}{N}\sum_{k=1}^{N} S_i^k$$

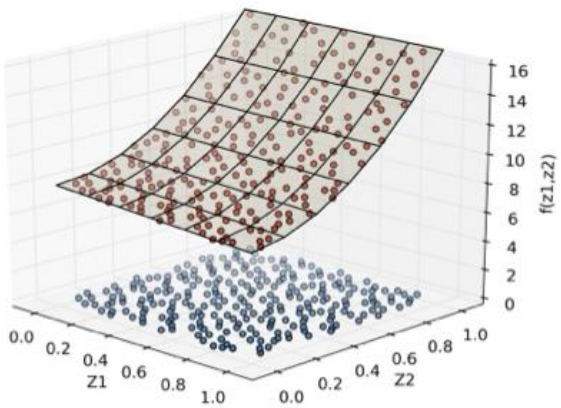
$$S_{i,weighted} = \frac{\sum_{k=1}^{N} S_i^k \cdot V_i^k}{\sum_{k=1}^{N} V_i^k}$$

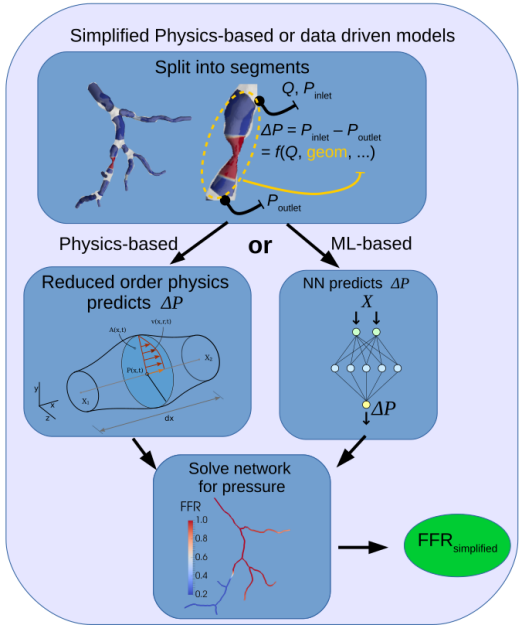


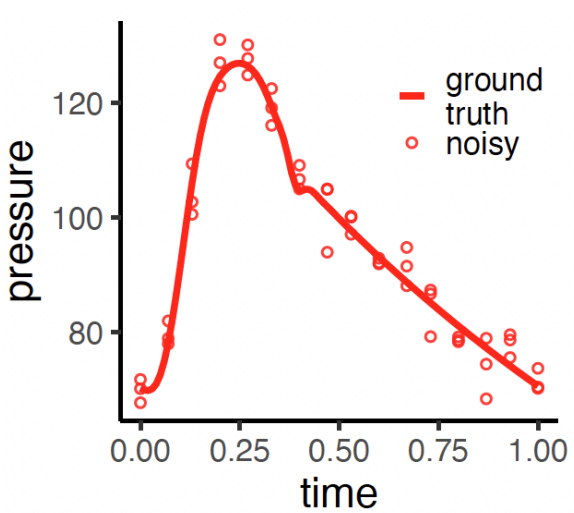


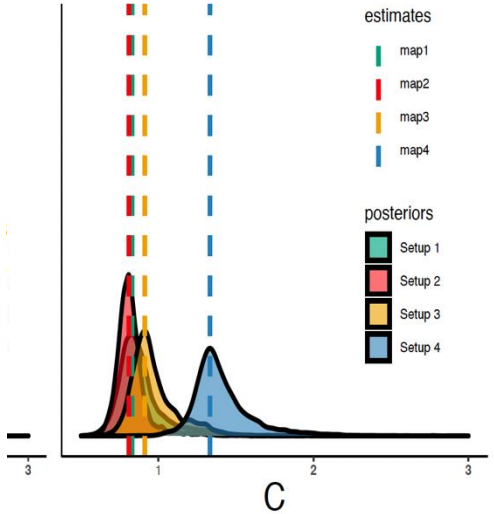
from ROMs and PCE to a broader surrogate landscape

- we have used ROMs and PCE as a surrogate for UQ and SA
- but other options exist:
  - GP good with small data
  - NN good for complex / high-dimensional problems
- different surrogates fit different situations
- the real question is:
  - “which surrogate, for which task, under which constraints?”









Matching surrogate to regime

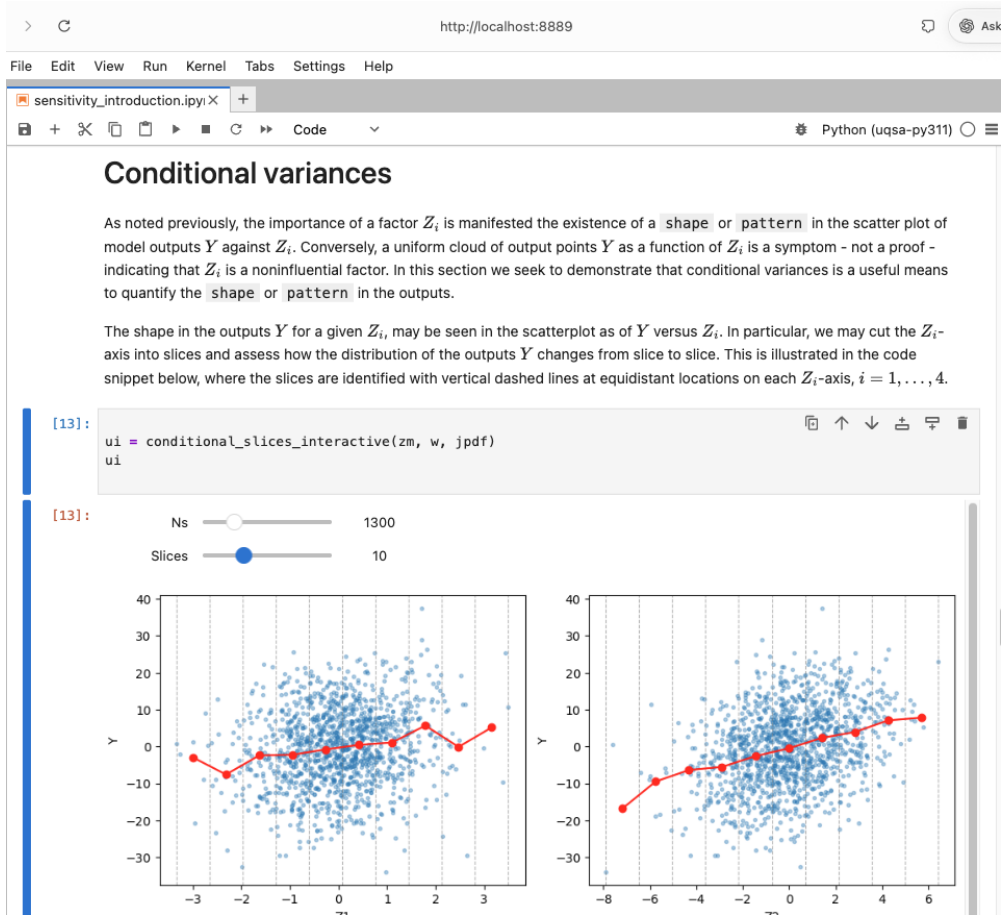
- small data/expensive simulations
  - PCE, GP
- more data
  - GP, NN
- complex / high-dimensional outputs
  - NN
- task matters:
  - UQ/SA PCE, GP
  - parameter estimation GP, NN
- no universally optimal surrogate

Explore UQSA further

Interactive notebooks and seminar material

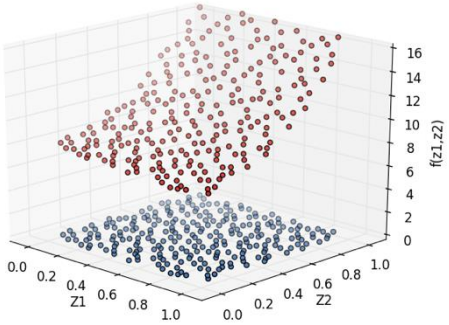
- Seminar material from Politecnic di Milano (Feb 2026)
- Focus:
  - building intuition for global sensitivity analysis
  - variance-based methods (Sobol)
  - Monte Carlo and polynomial chaos approaches
- Format:
  - short conceptual lectures
  - interactive Jupyter notebooks (Colab-ready)
- Access all seminar material

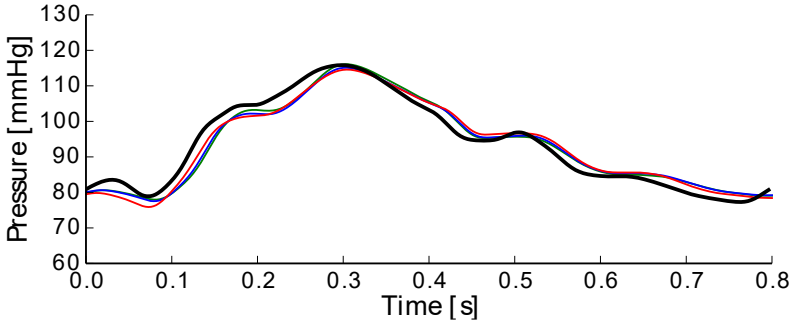
https://lrhgit.github.io/uqsa2025/

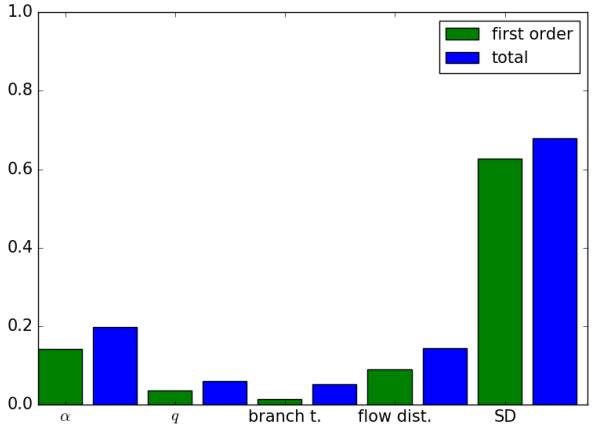


Takeaways

- Models are always uncertain — this uncertainty must be quantified
- Not all uncertainties matter equally
  - SA identifies what drives uncertainty
- Global SA enables:
  - prioritisation of parameters
  - model simplification
  - more robust decisions
- ROMs and surrogate models enable efficient UQSA
  - (PCE, GP, NN/ML)
- SA is not about everything — it is about what matters







Acknowledgements

This is a static version of the slides — some animations are omitted for clarity.

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